

Book :-

Mon Tue Wed Thu Fri Sat Sun

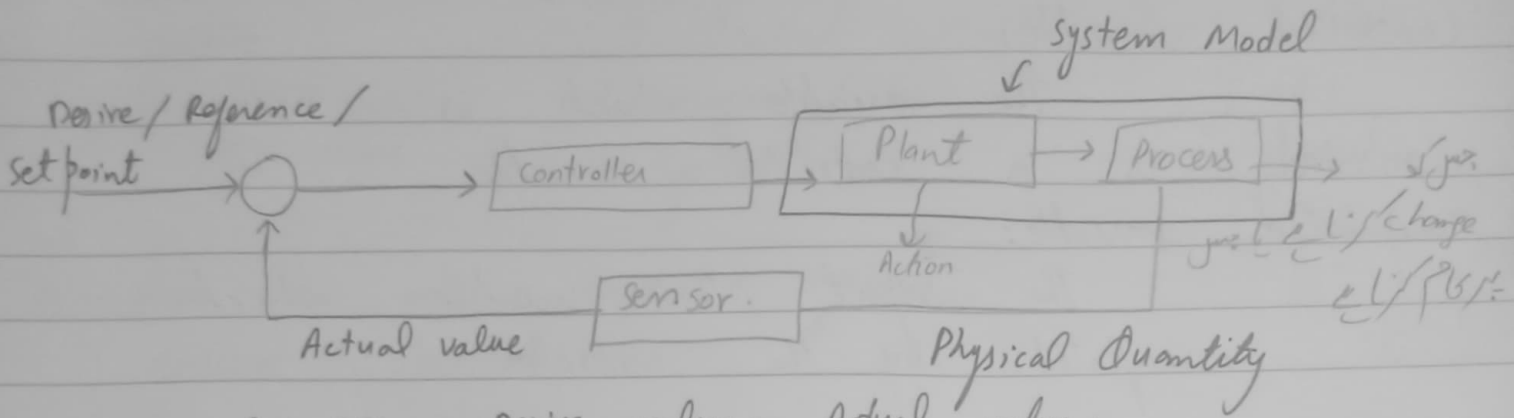
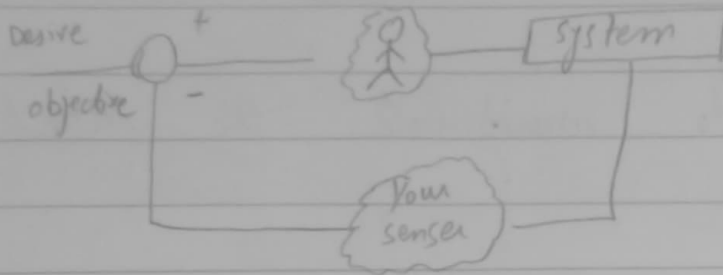
Control System Engineering

Day - 1



Define:-

Is a mechanism that we develop to manipulate the input and achieve a desired object.

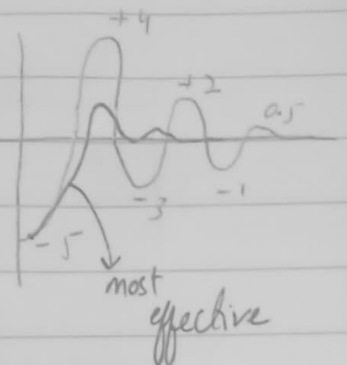


$$e = \text{error} = \text{Desire value} - \text{Actual value}.$$

Our main task is to zero the error.

Reference / Desire	Actual Value	error
25°C	30°C	-5°C
25°C	21°C	+4°C
25°C	28°C	-3°C
25°C	23°C	+2°C
25°C	26°C	-1°C
25°C	24.5°C	+0.5°C
25°C	25°C	0

∴ sampling time
1 min



Process: The operation that is being performed.

Plant: The machine / apparatus used to perform the operation

System: The whole environment

Sensor: The device which measures the physical quantity

Controller: The device used to manipulate the input signal to the plant

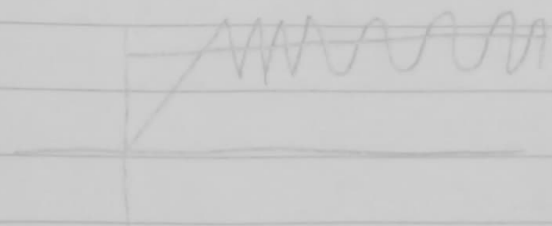
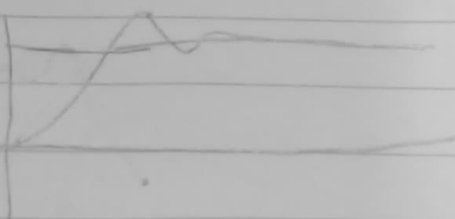
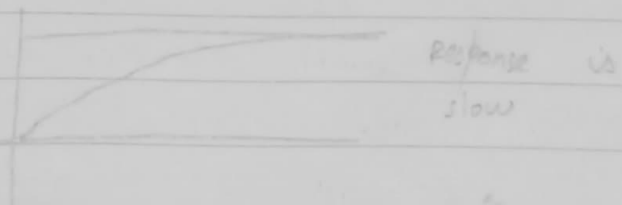
Controlled variable:

The variable which is being controlled is a controlled variable

Manipulated variable:

The input, the variable which is manipulated by the controller.

Elevator control (Significance of time for response ^{curves} control)

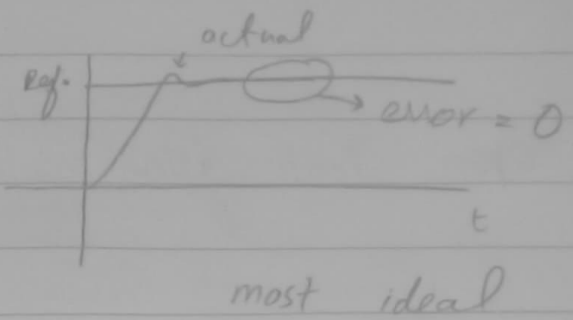


fastest but not ideal

OS $\rightarrow$ overshoot
 t_r = Rise time
 t_s = settle time
 e_{ss} = steady state error

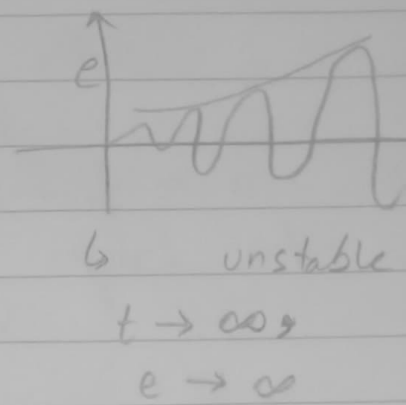
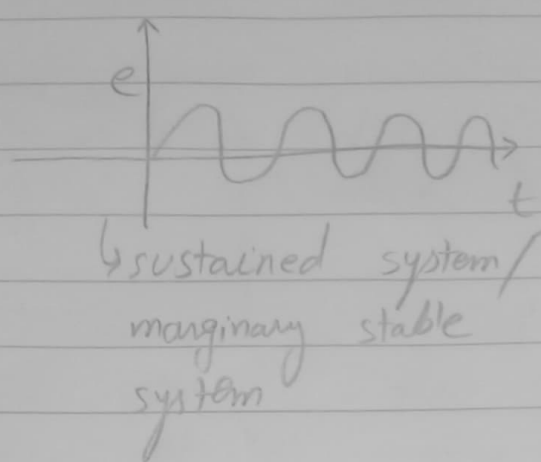
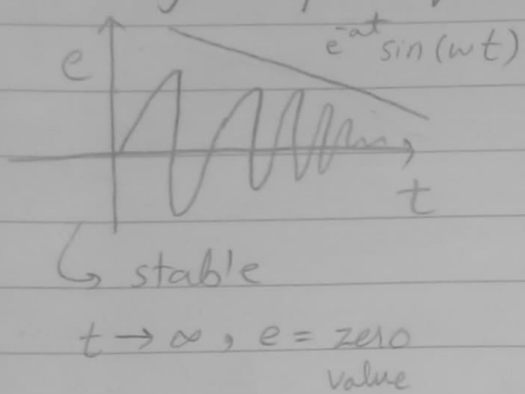


error : offset



most ideal

Stability of system :-



Objective to make Feedback system :-

- i) ensure stability
- ii) error minimization
- iii) minimize the overshoot, rise time, settling time, e_{ss}

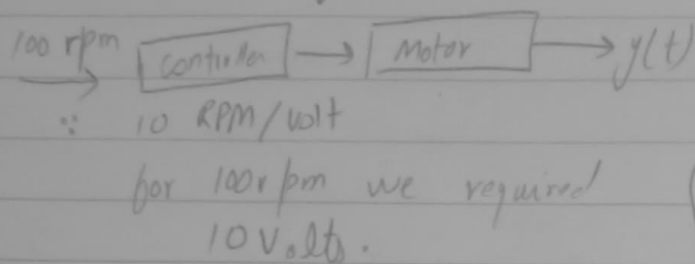
$\rightarrow$ Transient response and steady state are directly proportional to each other.

$\rightarrow t_r$ will always less than t_s .

Day 2

Open loop system

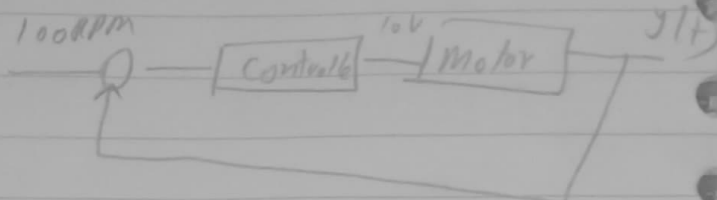
scenario:-



There is no feedback system in this loop.

Closed loop system

∴ we can't consider the distraction due to the environment. (for both)



- when load is applied the speed is slow so the controller is fed back with the error value.
- Feedback has a ability that it automatically remove its error.

Response :-

⇒ Total Response of a system = Natural Response + Forced Response

↳ depends on the dynamics of the system structure and their components

↳ depends on the type of the input applied.

$R(t) \rightarrow [h(t)] \rightarrow C(t) \rightarrow$ in time domain

$R(s) \rightarrow [H(s)] \rightarrow C(s) \rightarrow$ in s domain

Air war $\rightarrow$ air force system.



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Input

2

Output

$$3 \sin(\omega t) + 2$$

natural response

force response

$\therefore$ due to input

is constant in this case

the force response will be also constant cause of input nature.

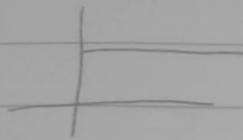
If there is a dynamic system it represent by the differential equation (d/dt). And it can be controlled.

Step inputs

impulse



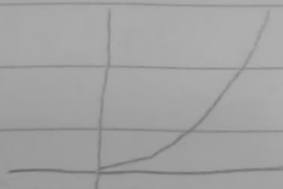
step



RAMP



parabolic



In mat lab we give these inputs to our system to check is it correct or not.

traits

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Date: / /

Control system Design Process:

- i) Defining various parts of physical into mathematical term
- ii) Draw a block diagram
- iii) Derive a mathematic model (in freq/ time domain) ← Laplace transfer function
- iv) Reduce the block diagram for simplicity
- v) Analyze the traits of a system (stability, os_r , t_r , t_p , en)
- 6) Design the system

⇒ Our reduced block diagram will be



